

TimeBase Extension for Load Forecasting using Informer and PMFormer

Prasun Baranwal

July 2026

Abstract

This paper presents an extension of the TimeBase framework for household load forecasting using Informer and PMFormer based architectures. The experiments were conducted on the *Load House 1.csv* dataset and evaluated for 1-hour, 8-hour, 16-hour, and 24-hour forecasting horizons. The work focuses on analysing training and validation loss behaviour over five epochs and comparing the learning pattern of TimeBase, TimeBase+Informer, and TimeBase+PMFormer.

1 Introduction

Accurate household load forecasting is important for energy planning, demand management, and smart grid optimisation. Load data contains temporal variations, non-linear behaviour, and changing consumption patterns, which makes forecasting challenging. Time-series forecasting models must therefore capture both short-term and long-term dependencies effectively.

In this work, the training was performed on the *Load House 1.csv* dataset. The study compares the learning behaviour of TimeBase with two extended models: TimeBase+Informer and TimeBase+PMFormer. The models were evaluated over 5 epochs for 1-hour, 8-hour, 16-hour, and 24-hour horizons.

2 Dataset

The dataset used in this work is *Load House 1.csv*. It contains household load values arranged as a time-series sequence. The same dataset was used for all model variants to maintain consistency in training and evaluation.

3 Baseline Results

The initial baseline comparison was performed using Autoformer, PatchTST, FEDFormer, and Informer. The models were evaluated using Mean Squared Error (MSE) and Mean Absolute Error (MAE).

Table 1: Baseline forecasting results on Load House 1.csv

Model	MSE 1h	MAE 1h	MSE 8h	MAE 8h	MSE 16h	MAE 16h	MSE 24h	MAE 24h
Autoformer	0.203187	0.287864	0.388144	0.341351	0.399429	0.366184	0.400078	0.370903
PatchTST	0.205795	0.270914	0.340980	0.379297	0.353017	0.386953	0.367246	0.403922
FEDFormer	0.230688	0.296629	0.343934	0.382666	0.371170	0.407577	0.380427	0.406446
Informer	0.181914	0.256252	0.293245	0.340351	0.330628	0.367206	0.405522	0.391100

4 TimeBase Extension

The proposed extension combines TimeBase with Informer and PMFormer. TimeBase provides a temporal modelling base, while Informer improves long-sequence dependency learning. PMFormer further strengthens temporal pattern modelling and can help the model capture more complex variations in load behaviour.

TimeBase+Informer is important because Informer is designed for efficient long-sequence forecasting. This makes it useful when the forecasting horizon increases from 1 hour to 24 hours. TimeBase+PMFormer is important because it enhances temporal representation and provides another strong extension for learning complex household load patterns.

5 Training and Validation Loss Values

Table 2 shows the training and validation loss values used for the 5-epoch analysis. These values correspond to the plotted loss curves for TimeBase, TimeBase+Informer, and TimeBase+PMFormer.

Table 2: Training and validation loss values for 5 epochs

Model	Horizon	Train E1	Train E2	Train E3	Train E4	Train E5	Val E1	Val E2	Val E3	Val E4	Val E5
TimeBase	1h	0.7200	0.5693	0.5029	0.4737	0.4608	0.3300	0.1893	0.1461	0.1328	0.1287
TimeBase	8h	0.7500	0.5191	0.4407	0.4140	0.4050	0.3300	0.2130	0.1716	0.1570	0.1518
TimeBase	16h	0.7800	0.4561	0.3712	0.3490	0.3432	0.3200	0.2292	0.1930	0.1785	0.1728
TimeBase	24h	0.8000	0.3758	0.2919	0.2753	0.2720	0.3200	0.2493	0.2182	0.2044	0.1984
TimeBase+Informer	1h	0.6700	0.5198	0.4536	0.4245	0.4116	0.3100	0.1728	0.1306	0.1176	0.1136
TimeBase+Informer	8h	0.6900	0.4676	0.3921	0.3664	0.3577	0.3100	0.1944	0.1536	0.1391	0.1340
TimeBase+Informer	16h	0.7100	0.4059	0.3262	0.3054	0.2999	0.3000	0.2089	0.1725	0.1580	0.1523
TimeBase+Informer	24h	0.7200	0.3304	0.2533	0.2380	0.2350	0.2900	0.2197	0.1887	0.1750	0.1690
TimeBase+PMFormer	1h	0.6600	0.5022	0.4327	0.4021	0.3886	0.3000	0.1639	0.1221	0.1092	0.1053
TimeBase+PMFormer	8h	0.6900	0.4576	0.3787	0.3519	0.3428	0.3100	0.1907	0.1485	0.1336	0.1283
TimeBase+PMFormer	16h	0.7100	0.3966	0.3145	0.2930	0.2874	0.3000	0.2049	0.1669	0.1518	0.1458
TimeBase+PMFormer	24h	0.7200	0.3225	0.2439	0.2283	0.2252	0.2900	0.2155	0.1826	0.1682	0.1618

6 Result Graphs

7 Discussion

The loss values show a decreasing trend over five epochs for all three model categories. TimeBase acts as the base temporal forecasting model. TimeBase+Informer starts with lower loss values than TimeBase and continues to reduce both training and validation loss across epochs. This indicates that the Informer extension can support better temporal dependency learning.

TimeBase+PMFormer also shows a consistent reduction in training and validation loss. Compared with the base TimeBase model, the PMFormer extension provides lower loss values across the considered horizons. This suggests that PMFormer can improve the representation of complex temporal load patterns, especially for longer forecasting horizons.

8 Conclusion

This work presents TimeBase extensions for household load forecasting using Informer and PMFormer. The experiments were performed using *Load House 1.csv* and evaluated across 1-hour, 8-hour, 16-hour, and 24-hour forecasting horizons. The results show that TimeBase+Informer and TimeBase+PMFormer are useful extensions for improving temporal learning and reducing training and validation loss over five epochs.